1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

In NMIBC patients treated with intravesical BCG, tumors with TLS-rich immune microenvironments will have higher therapy_response than TLS-poor tumors, but this benefit is contingent on the tumor's MLH1 status and TWIST1 promoter methylation. Specifically, TLS-rich tumors that harbor MLH1 loss (dMMR/MSI) in combination with high TWIST1 promoter methylation will show significantly reduced therapy_response (and shorter recurrence-free survival) compared with TLS-rich tumors lacking this MLH1/TWIST1 axis. The proposed interaction suggests that MLH1-dMMR–driven neoantigenicity may be offset by EMT/stromal remodeling signaled by TWIST1 methylation, leading to immune exclusion or dysfunction that decouples TLS from effective anti-tumor immunity. Thus, TLS presence predicts BCG benefit primarily in MLH1-intact and/or low TWIST1 methylation tumors; the MLH1-dMMR plus high TWIST1 methylation axis defines a biomarker-defined subset with poor response who may require alternative or combination immunomodulatory strategies.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	MLH1, TWIST1
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

MLH1: MLH1 as representative MMR gene) — provides a genomic axis that modulates neoantigen load and immune recognition; (c) urine DNA methylation markers (TWIST1 as exemplar) — supplies an epigenetic/phenotypic axis that may drive immune exclusion or therapy resistance.

TWIST1: TWIST1 as exemplar) — supplies an epigenetic/phenotypic axis that may drive immune exclusion or therapy resistance.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	MSH2	Supporting modifier from knowledge graph
Secondary	MSH6	Supporting modifier from knowledge graph

Secondary	PMS2	Supporting modifier from knowledge graph
Secondary	CD8A	Supporting modifier from knowledge graph
Secondary	CD3D	Supporting modifier from knowledge graph
Secondary	CD274	Supporting modifier from knowledge graph
Secondary	TMB	Supporting modifier from knowledge graph
Secondary	SDC2	Supporting modifier from knowledge graph
Secondary	TFPI2	Supporting modifier from knowledge graph
Secondary	ApoA1/HDL	Supporting modifier from knowledge graph
Secondary	ADNP	Supporting modifier from knowledge graph
Secondary	PTGDS	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- MSH2: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- MSH6: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.